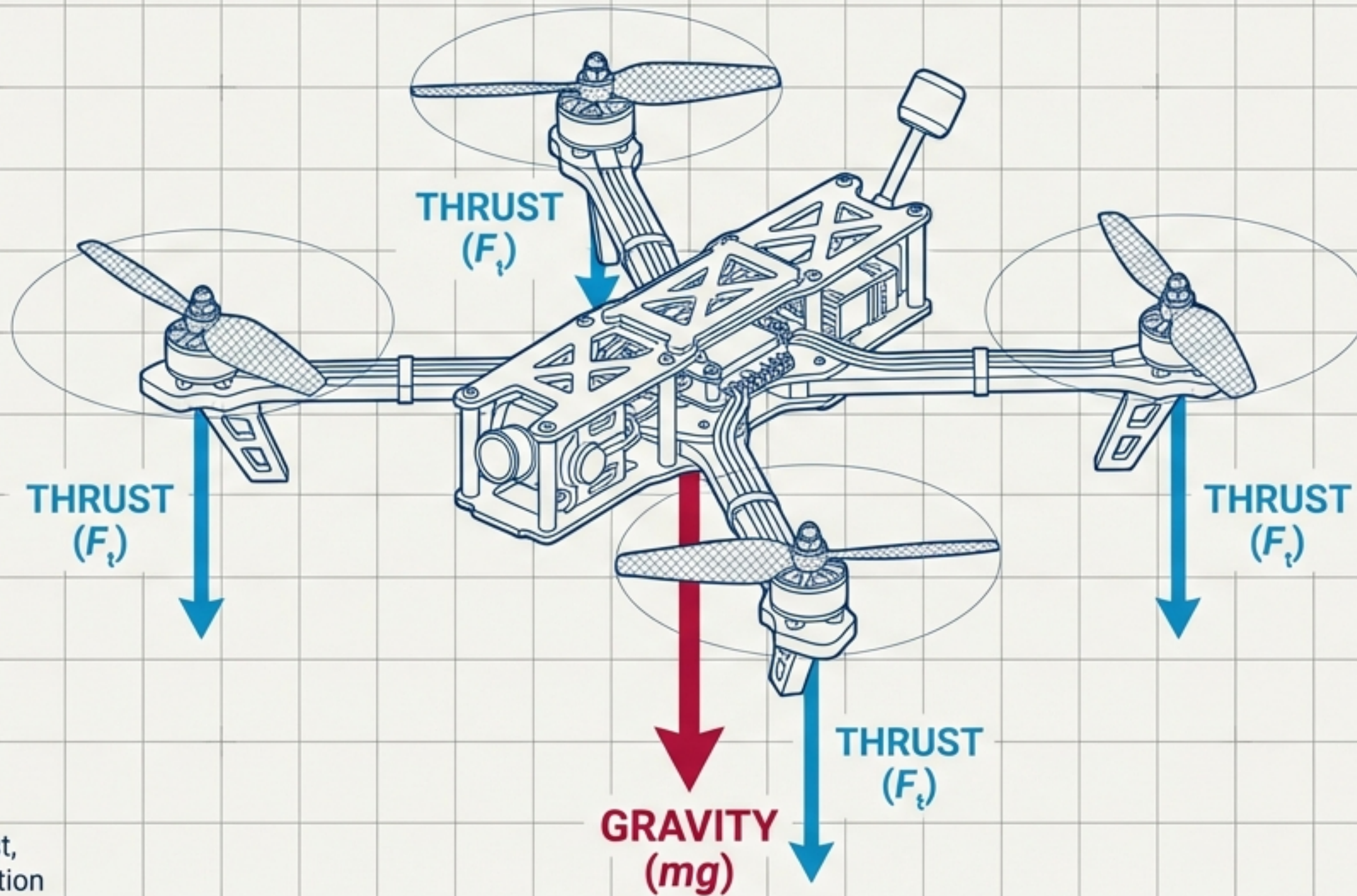


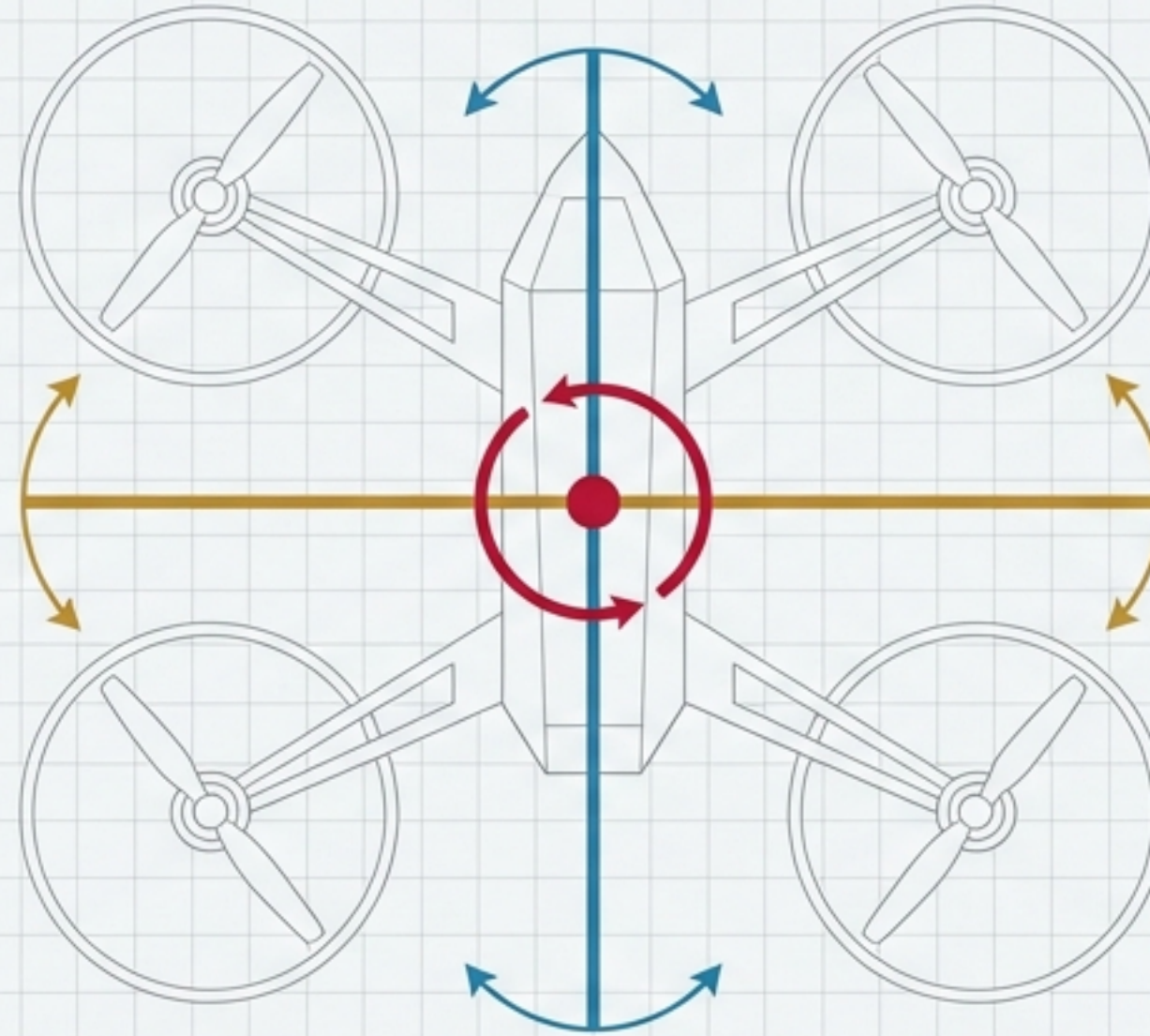
ENGINEERING THE SKY

The Aerodynamics and Physics of Quadcopter Flight



A Visual Framework for Thrust,
Drag, Lift, and Differential Motion

Defining the Three-Dimensional Flight Canvas



X-Axis: Roll Movement



(Left/Right tilt) - Controls the side-to-side tilting motion of the aircraft, allowing for lateral displacement and banking turns.

Y-Axis: Pitch Movement



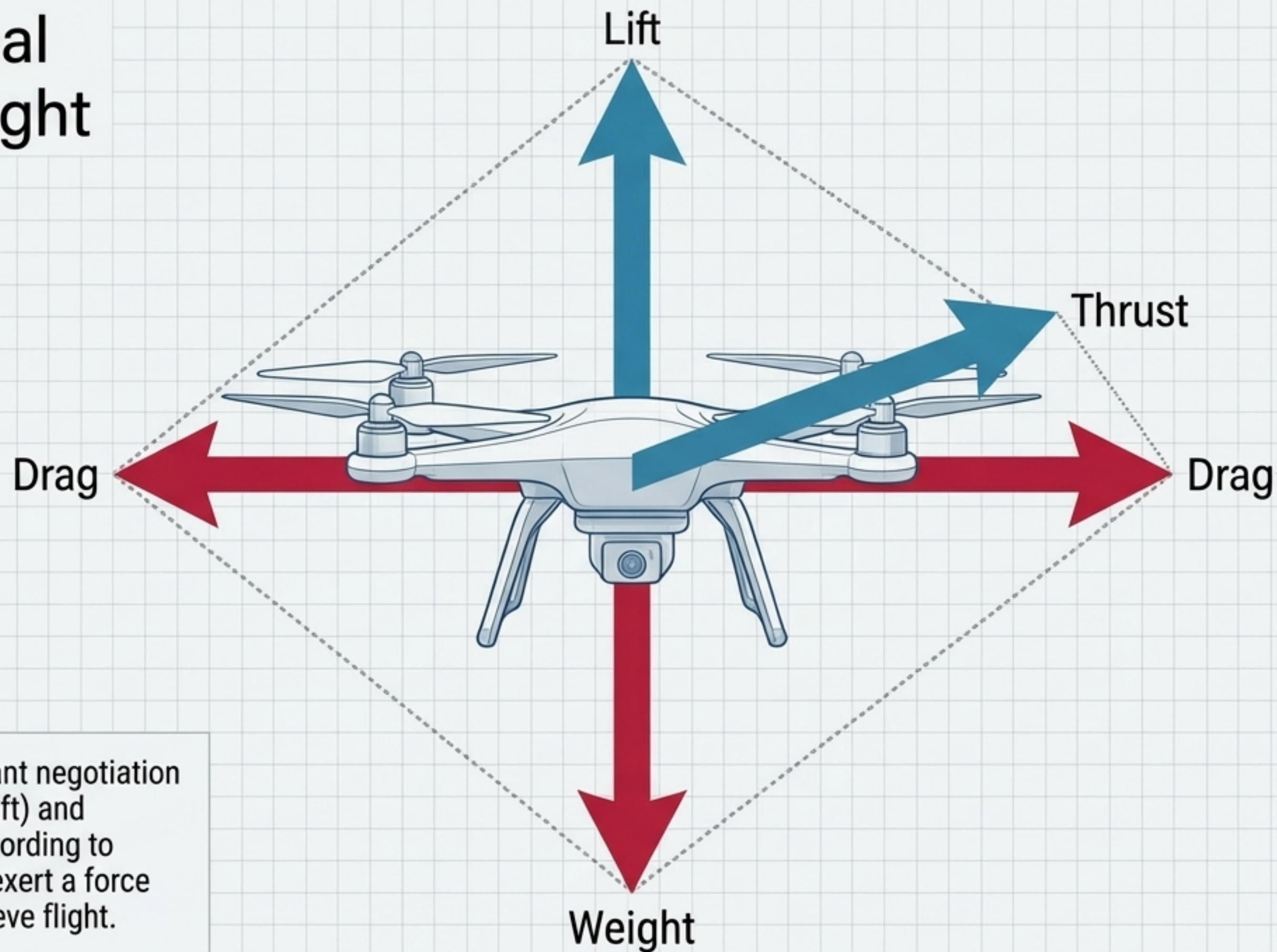
(Forward/Backward tilt) - Controls the nose-up and nose-down tilting motion of the aircraft, allowing it to move forward or backward and climb or descend.

Z-Axis: Yaw Movement



(Clockwise/Counter-Clockwise rotation) - Controls the rotation of the aircraft around its vertical axis, allowing it to change its heading without changing its flight path.

The Four Universal Constraints of Flight

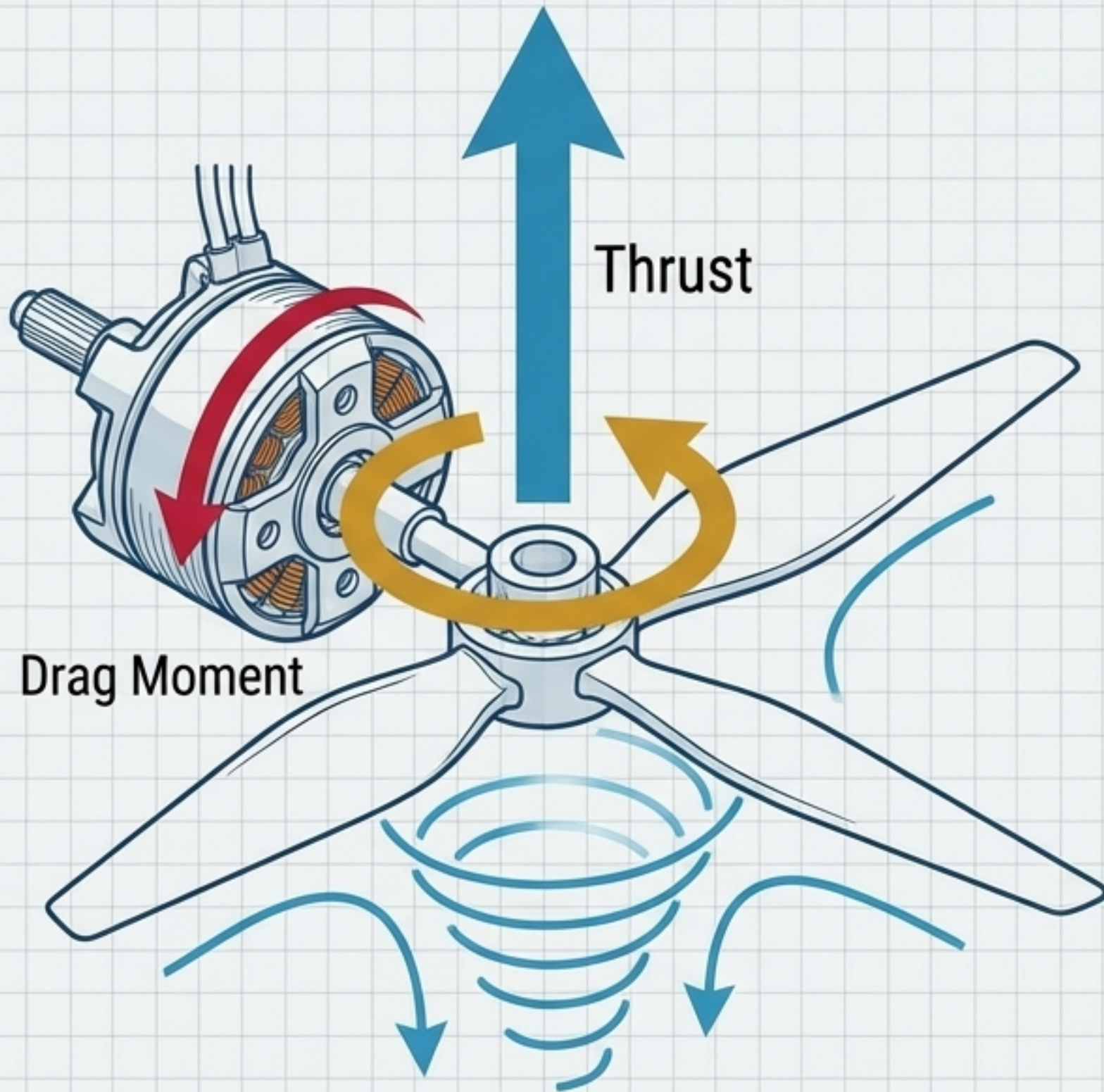


Every movement in the air is a constant negotiation between propulsion forces (Thrust/Lift) and resistance forces (Gravity/Drag). According to Newton's Third Law, the drone must exert a force equal and opposite to gravity to achieve flight.

The Four Forces Diagnostic Matrix

Thrust	Weight	Lift	Drag
Source: BLDC Motors & Propellers	Source: Earth's Gravity (9.8 m/s ²)	Source: Combined thrust vectors	Source: Air resistance
Direction: Action force opposing gravity	Direction: Pulls directly downward	Direction: Vertical	Direction: Opposite to flight path
Role: Primary propulsion. Proportional to the square of angular motor speed.	Role: The fundamental constraint. Total Thrust must be double the weight for safe maneuvers.	Role: The net upward force that pulls the mass into the air.	Role: Friction. The drone's cruise speed must be sufficient to overcome this force.

The Engine: Generating Thrust and Overcoming Drag



$$F_{\text{total}} = K_f \times \omega^2$$

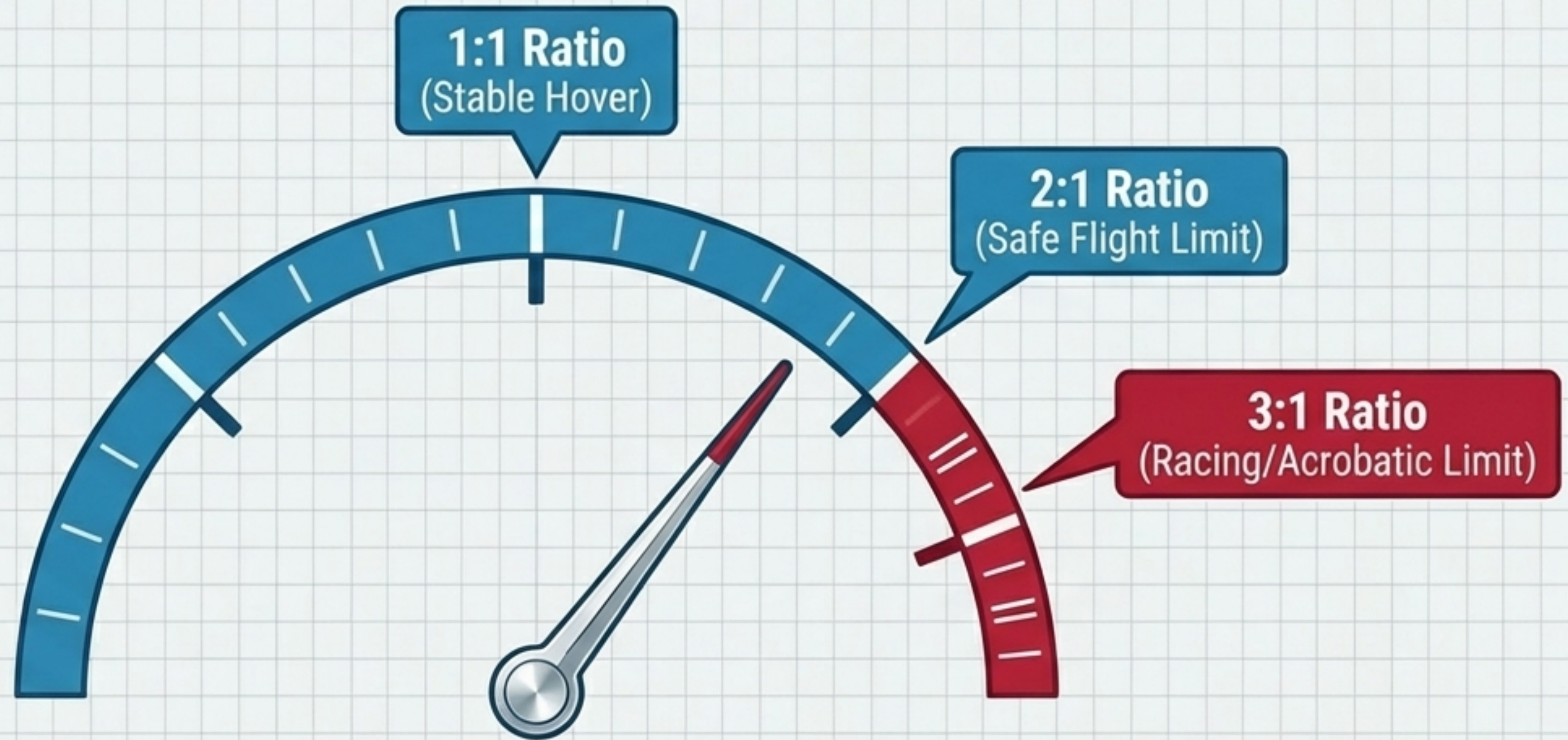
(Thrust is proportional to the square of angular speed)

Spinning propellers push air downward. Newton's Third Law dictates an equal upward reaction force. However, approximately $\frac{1}{4}$ of the total force produced by the rotor is consumed just overcoming the motor's own aerodynamic drag.

Therefore, $F_{\text{effective}} = \frac{1}{4} (\text{Total Thrust})$.

Thrust-to-Weight Ratio (TWR): The Ultimate Performance Metric

$$\text{TWR} = \frac{\text{Total Thrust at 100\% Throttle}}{\text{Weight of Drone}}$$

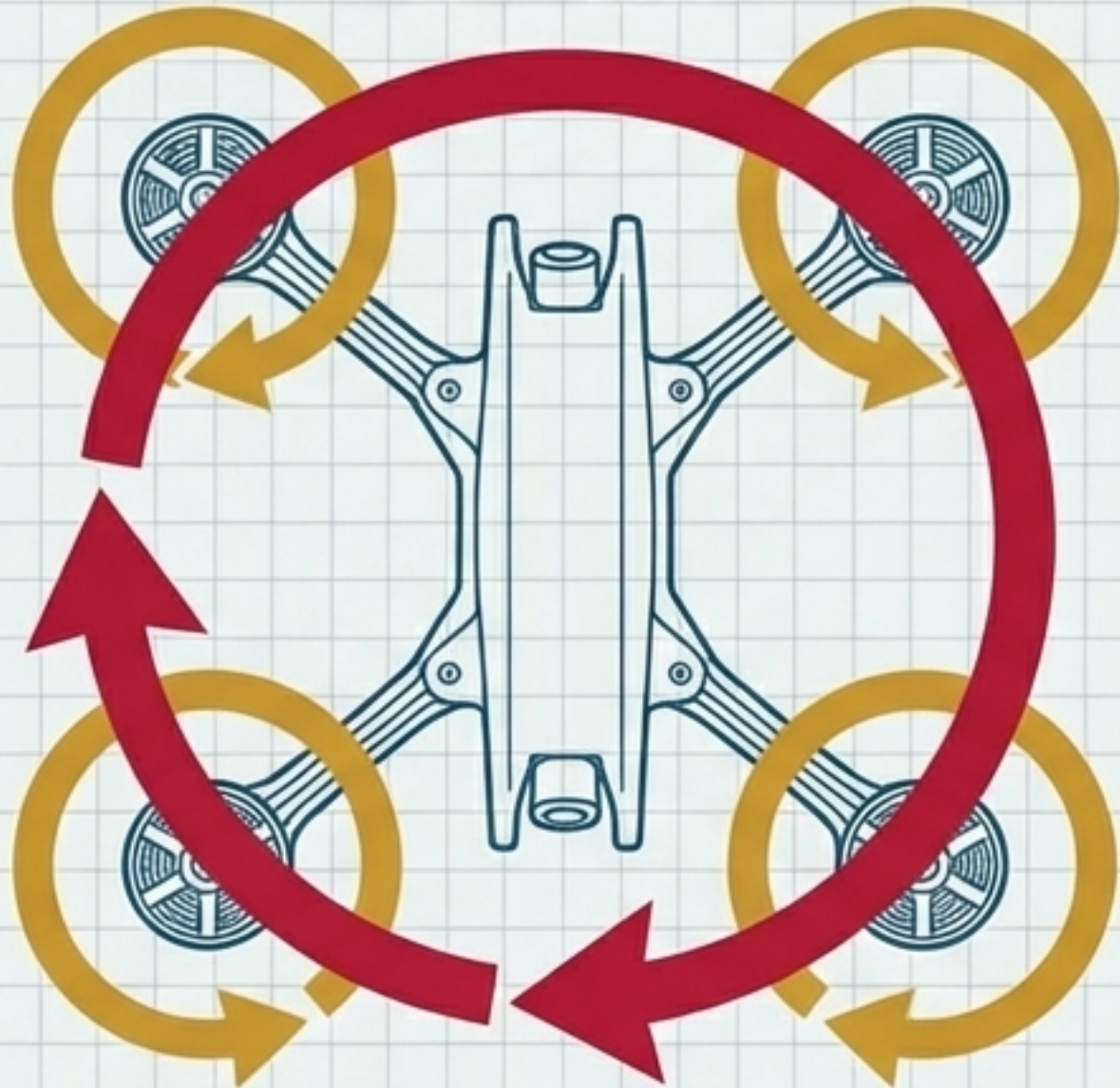


For a multirotor to hover stably, it must generate thrust exactly equal to its weight at 50% throttle. To safely cruise, maneuver, and combat wind, the total thrust capacity must be at least twice the drone's weight.

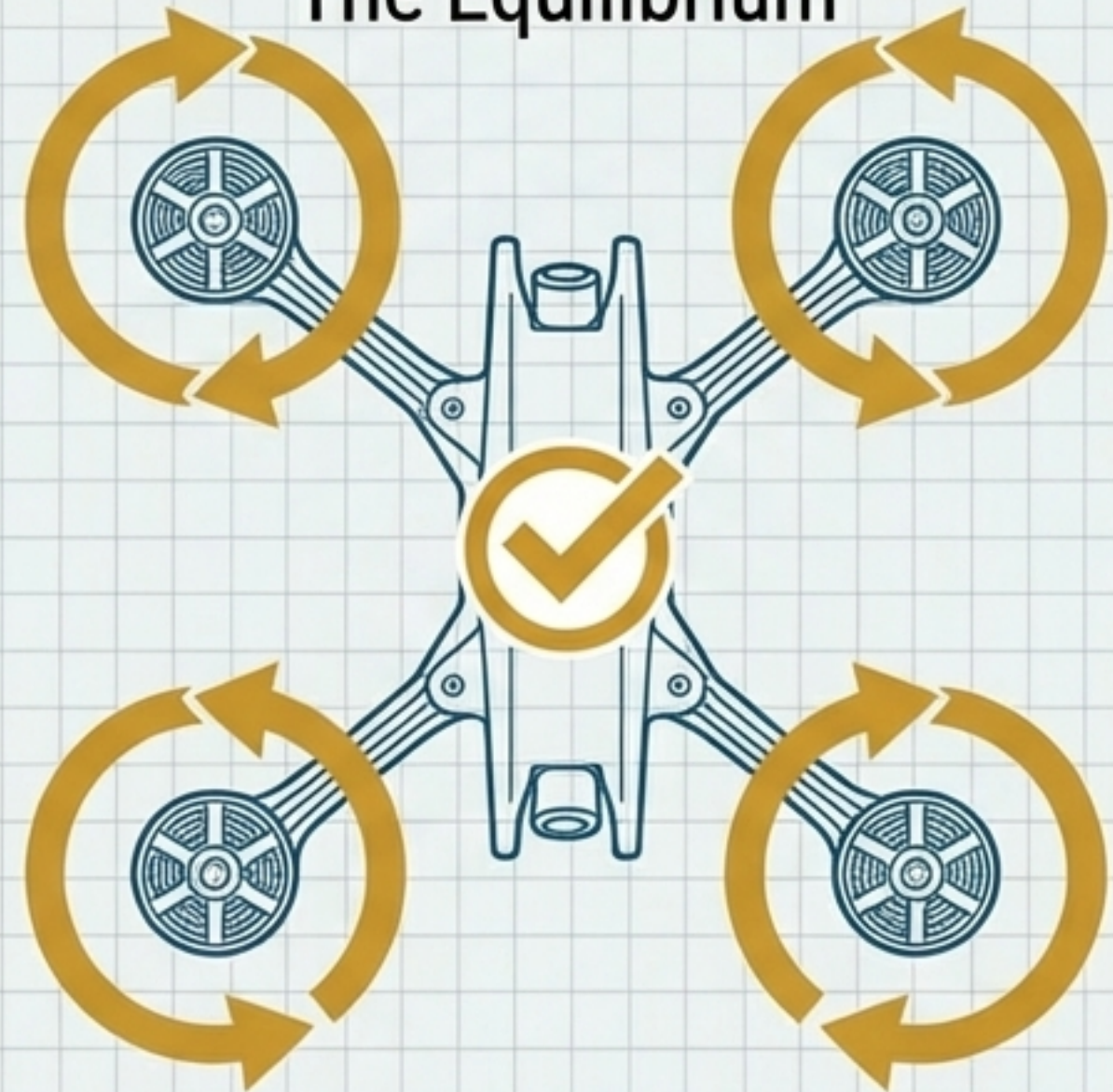
Newton's Third Law & The Anti-Torque Configuration

Every spinning propeller applies an equal and opposite rotational force (torque) to the drone's body. By alternating diagonal pairs—two clockwise and two counter-clockwise—the opposing **torques perfectly cancel each other out**, ensuring the drone does not spin out of control on its Z-axis.

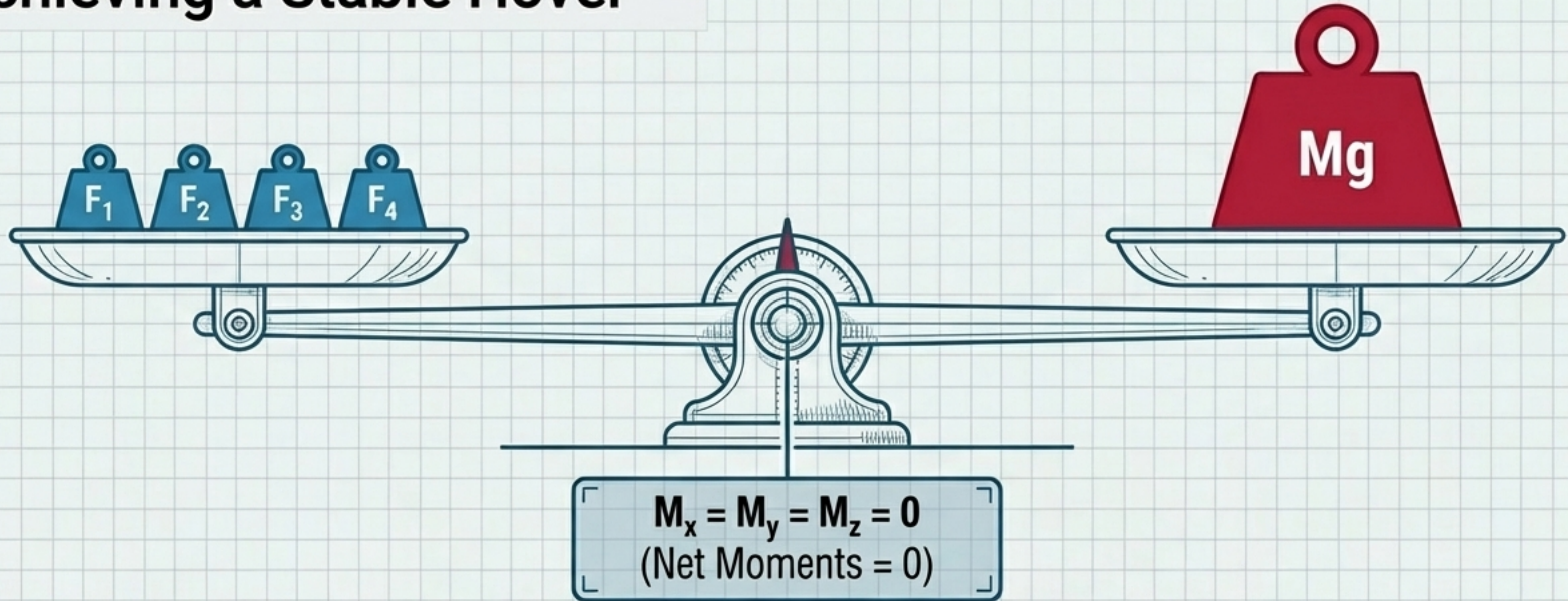
The Problem



The Equilibrium



The Physics of Equilibrium: Achieving a Stable Hover

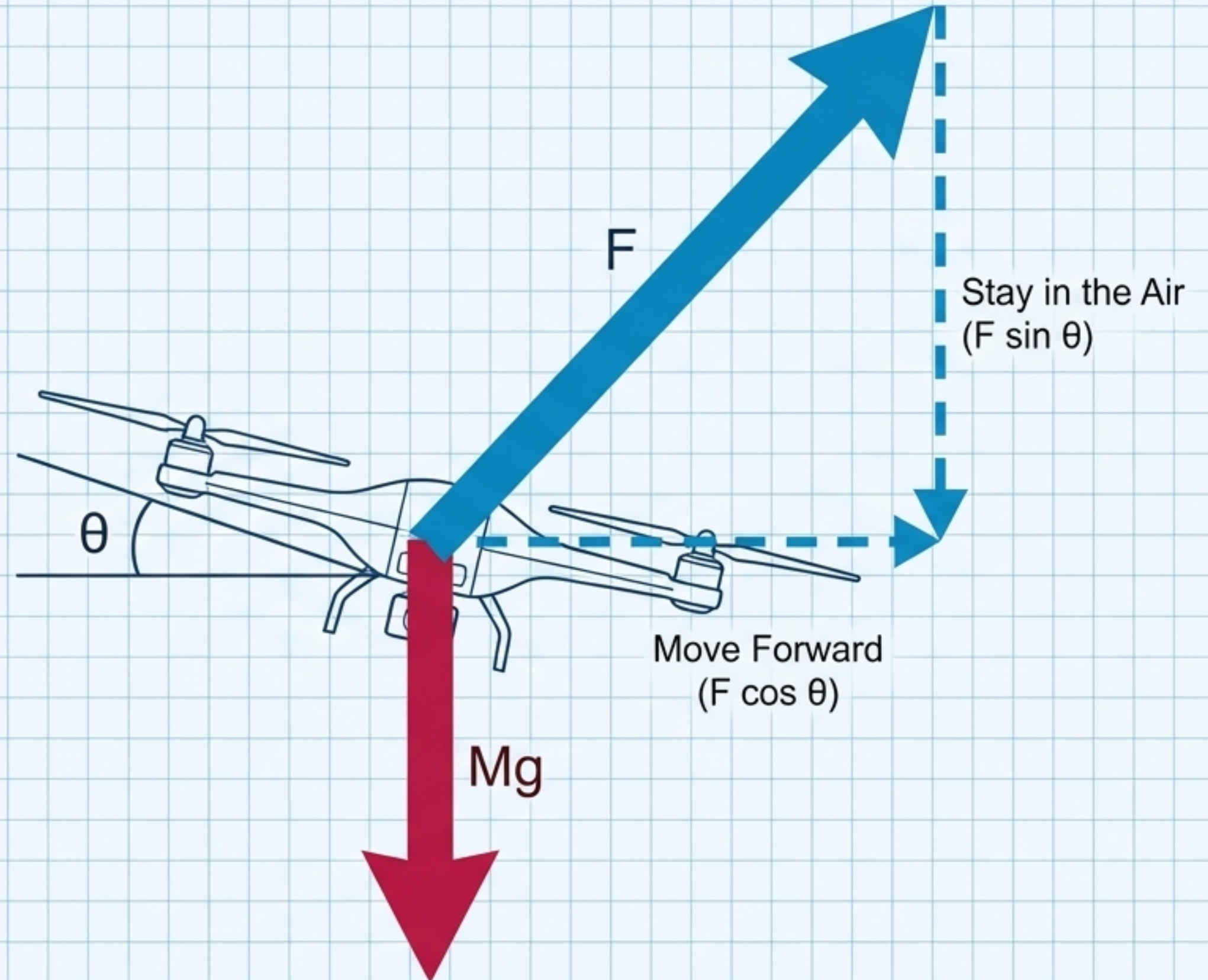


A stable hover is achieved only when the sum of the thrust from all four motors equals the downward force of gravity, and all rotational moments across the center of mass cancel out.

The Hover Condition: $F_{\text{total}} = F_1 + F_2 + F_3 + F_4 - Mg = 0$

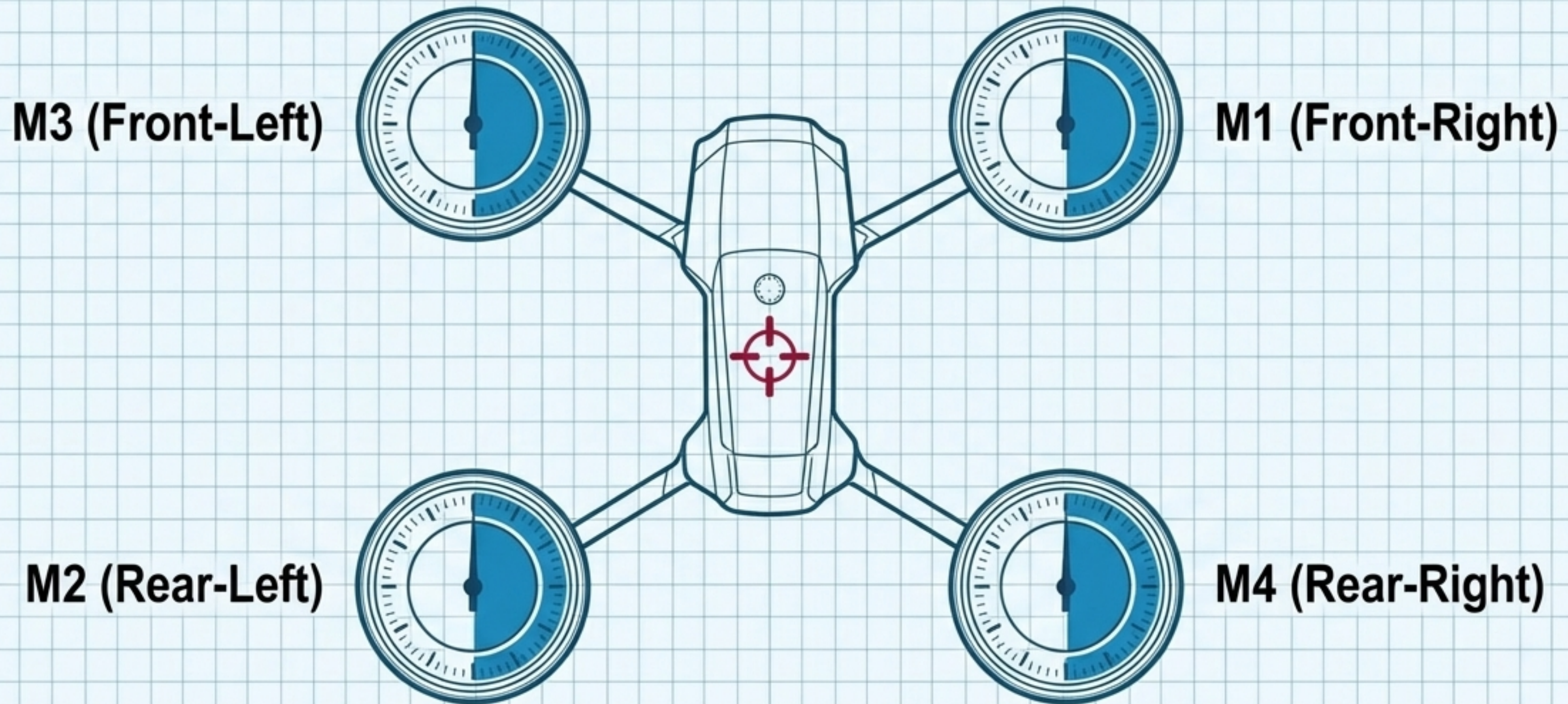
Vector Splitting: The Anatomy of Horizontal Motion

To move horizontally, the drone tilts. The total thrust vector splits. The vertical component battles gravity to maintain altitude, while the horizontal component accelerates the drone through space ($F = Ma$).

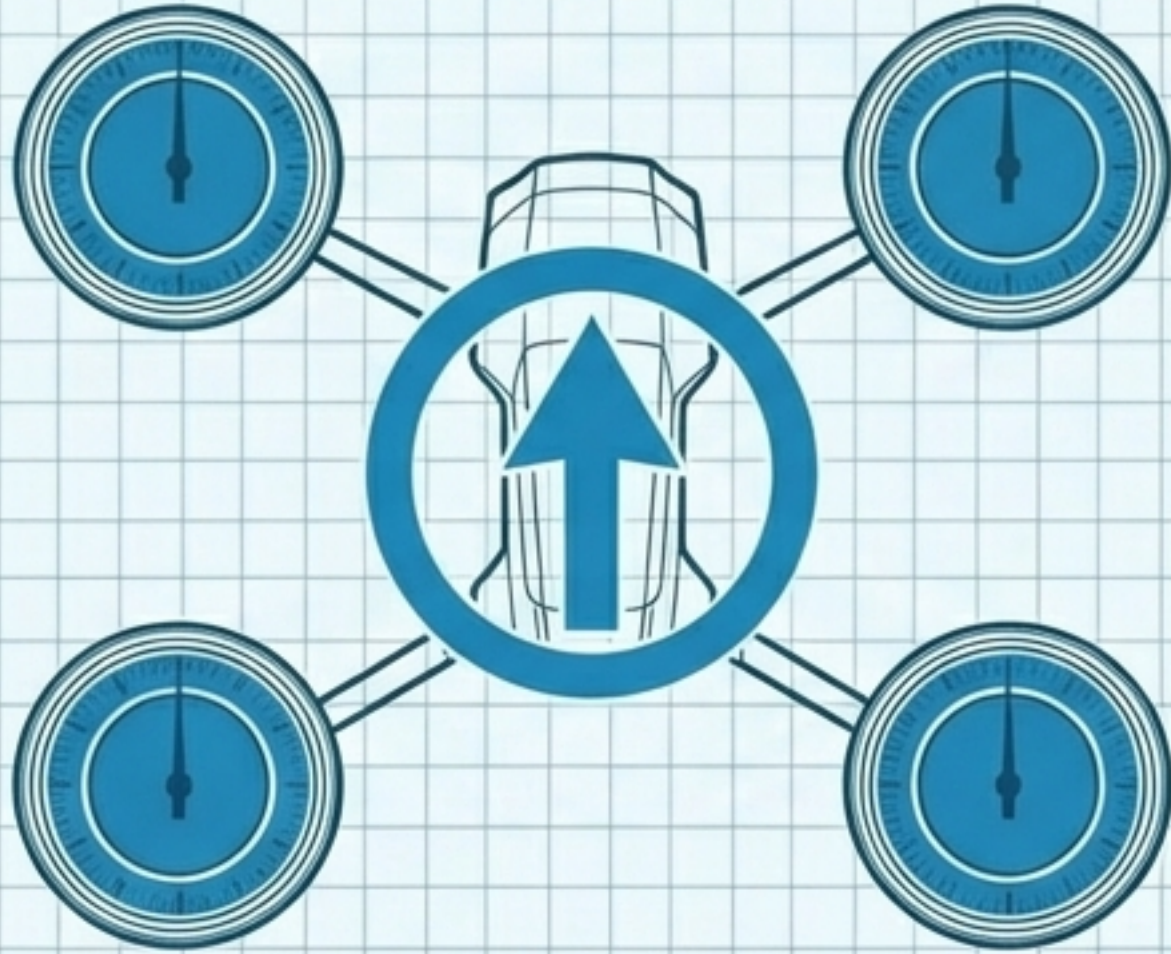


The Secret of Motion: Differential Thrust

Because quadcopters have no physical control surfaces (like airplane flaps or rudders), every single directional maneuver is achieved solely by altering the relative RPM (Revolutions Per Minute) of individual motors. Creating intentional imbalances in thrust is how a drone flies.

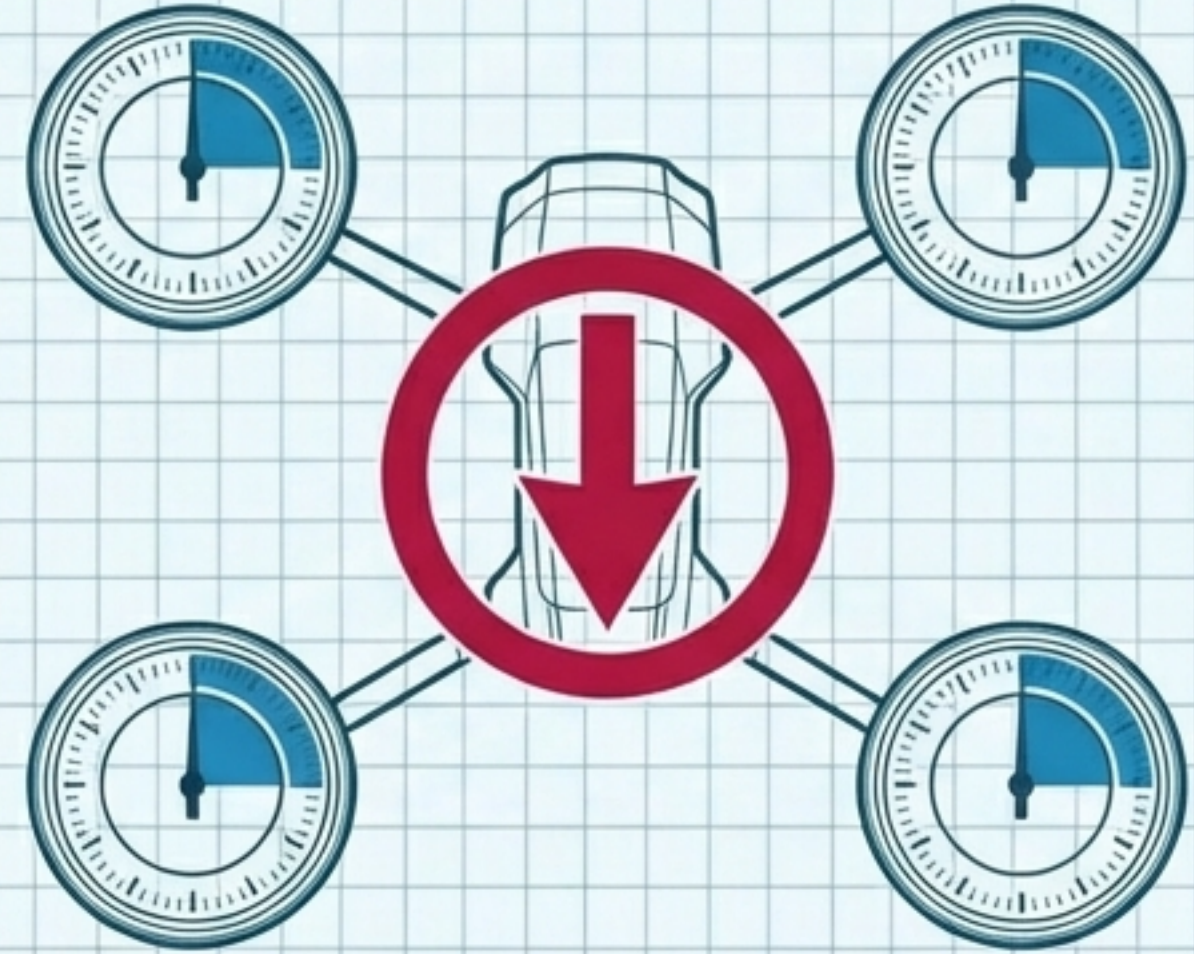


Vertical Maneuvers: Breaking Equilibrium



Ascent: Increasing throttle raises the RPM of all four rotors equally. Total thrust exceeds the drone's weight, creating upward acceleration.

$$F_{\text{total}} > Mg$$



Descent: Decreasing throttle drops the RPM of all four rotors equally. Gravity overcomes total thrust, pulling the system downward.

$$F_{\text{total}} < Mg$$

Pitch Maneuvers (Forward / Backward Motion)

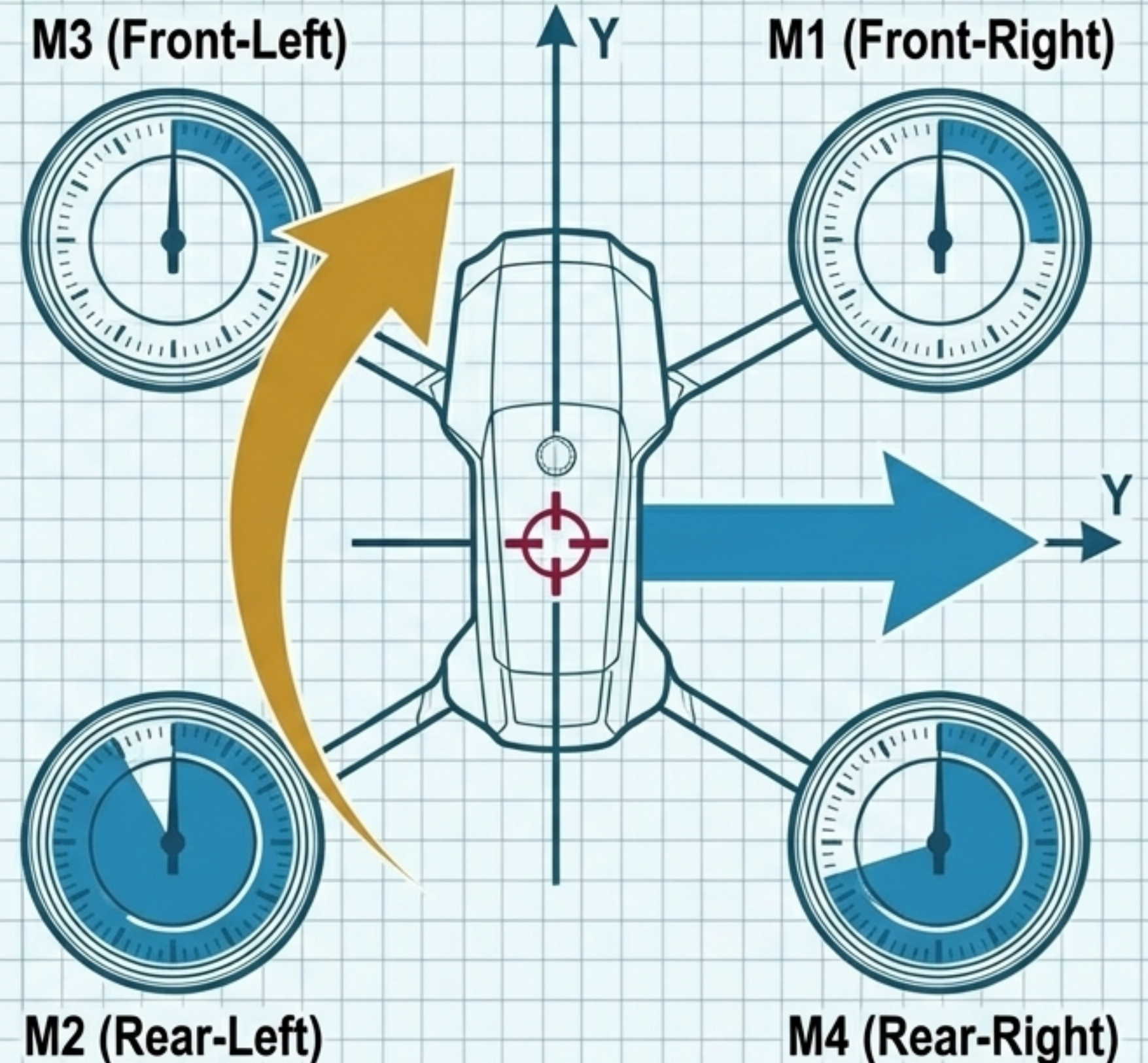
The Physics:

By speeding up the back motors and slowing down the front motors equally, the total upward thrust remains constant (maintaining altitude), but the moment across the center of gravity shifts.

The Formula:

Moment due to front motors <
Moment due to back motors
($F_{\text{front}} \times L < F_{\text{back}} \times L$).

The rear lifts, tilting the thrust vector forward.



Roll Maneuvers (Lateral Motion)

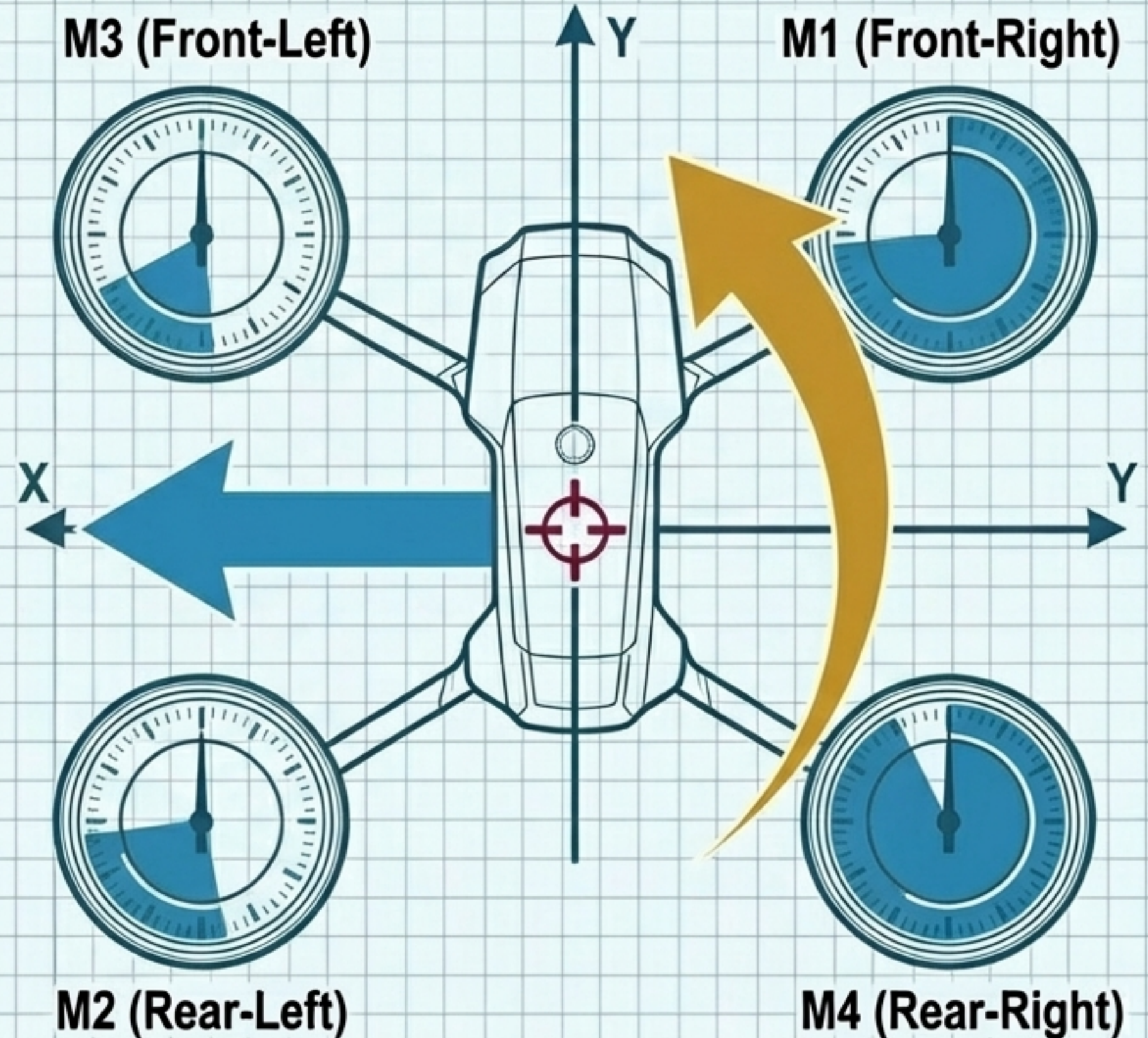
The Physics:

To move left, the two motors on the right side rotate faster, while the two on the left slow down. Total lift remains identical, but the force imbalance pushes the drone left.

The Formula:

Moment due to left motors <
Moment due to right motors
($F_{\text{left}} \times L < F_{\text{right}} \times L$).

The right side lifts, tilting the thrust vector to the left.



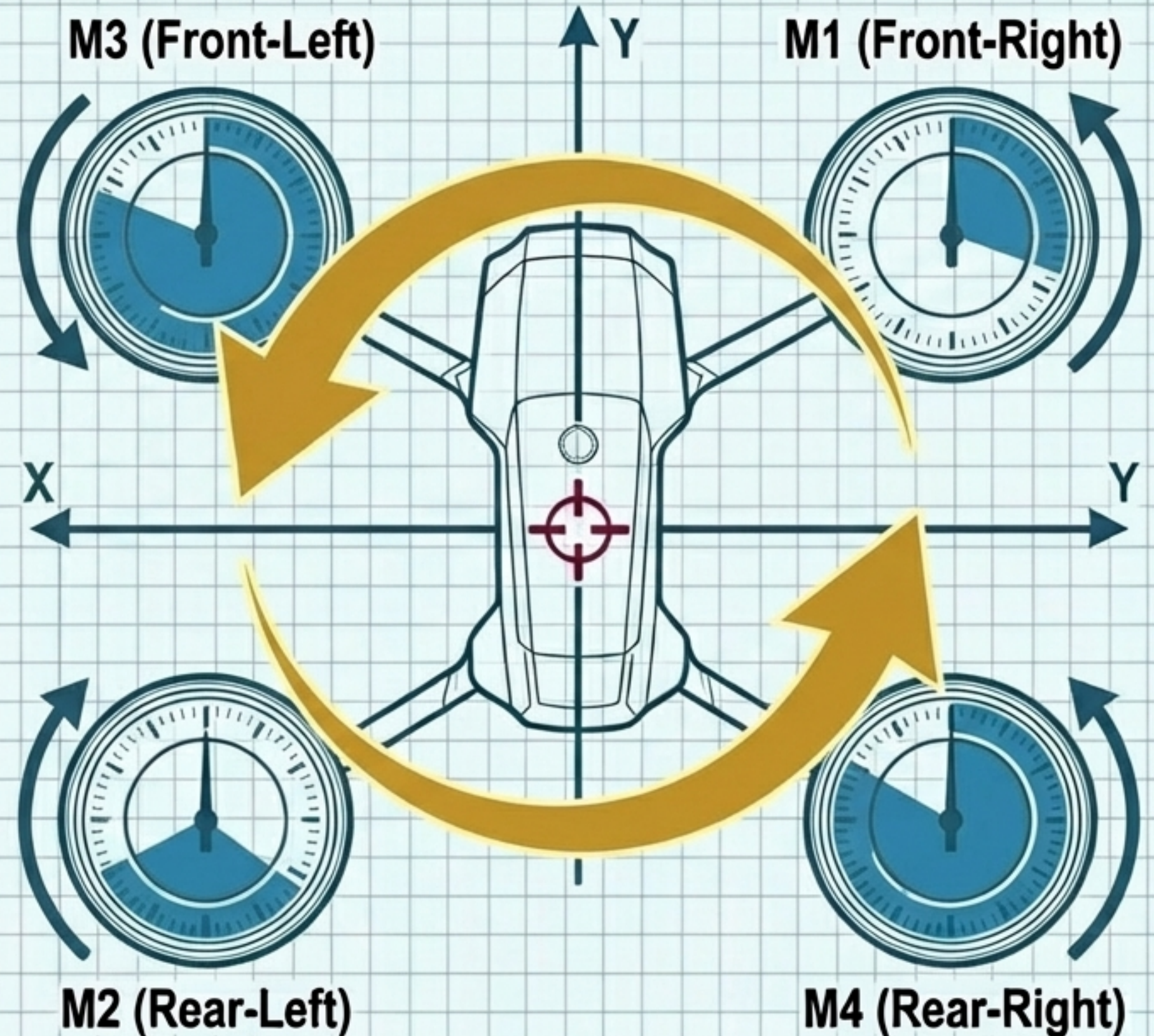
Yaw Maneuvers (Z-Axis Rotation)

The Physics:

Yaw does not require tilting. Instead, it weaponizes Newton's Third Law. To rotate clockwise, the drone slows down the CW motors and speeds up the CCW motors.

The Result:

The reaction force of the CCW motors is now greater than the CW motors. The unbalanced net torque physically forces the airframe to spin clockwise.



The Synthesis:

Differential Thrust Master Matrix

	Front-L (M3)	Front-R (M1)	Rear-L (M2)	Rear-R (M4)	Result
Hover	=	=	=	=	Thrust = Weight, Net Torque = 0
Ascent	UP ▲	UP ▲	UP ▲	UP ▲	Thrust > Weight
Pitch Fwd	DOWN ▼	DOWN ▼	UP ▲	UP ▲	Tilt Forward (Y-Axis)
Roll Left	DOWN ▼	UP ▲	DOWN ▼	UP ▲	Tilt Left (X-Axis)
Yaw CW	CCW Motors: UP ▲		CW Motors: DOWN ▼		Net Torque spins drone Clockwise

By orchestrating thousands of micro-adjustments to these four vectors per second, the flight controller transforms rigid physics into fluid, three-dimensional flight.